W-A-450/14
JANUARY 3, 1992

FEDERAL SPECIFICATION

DETECTOR, CABLE, FLEXIBLE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit and type. This specification covers Unit Number 1, Type XIV, Flexible Cable Detector.

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, of the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

3. REQUIREMENTS.

3.1 Qualification. The flexible cable detector furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been tested and shall have passed the qualification tests specified in 4.1.

3.2 Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of operating within the temperature and humidity ranges indicated below:

0 degrees C to +49 degrees C and relative humidity of 0% to 85%, non-condensing.

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FSC 6350

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All units shall withstand the tests in 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. All equipment supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 10 000 hours.

3.4 Operation. The flexible cable detector shall consist of a cable sensing element and a processor. When properly installed on a building's interior surface, the detector shall detect an intrusion attempt.

This detector can be used on the following surface types: walls, ceilings, roofs, floors, pipes, ducts, and metallic screening. The cable shall be designed to attach directly to the surface or be installed in conduit attached to the surface to be protected.

The processor shall use the electrical signal produced by the minute flexing of the cable during a penetration attempt to sense an intrusion and to operate an alarm relay. The processor shall also contain the supervisory alarm relay circuitry to detect a cut or a shorted sensing cable.

The system shall be designed to monitor a high or low frequency band (selected by user) depending on the type of intrusion detection required. Detection of breaking penetration attempts shall require the selection of the low frequency band. The high frequency band must be selected to detect cutting, drilling, or chipping penetration attempts.

Input voltage shall be 8 to 18 volts DC, reverse polarity protected, or 10 to 28 volts AC RMS, 60 cps. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment.

Alarm and tamper switch impedance shall be as follows:

Open state:
- greater than 100 000 ohms.

Closed state:
- For solid state relays - 50 ohms or less.
- For mechanical relays - 0.1 ohms or less.

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parts and the enclosure sufficiently displaced from all internal

The processor shall be able to monitor from 3000 mm up to 610 000 mm of sensor cable with uniform sensitivity. The sensitivity shall be adjustable to any of ten discrete levels. The system shall also allow the selection of the number of impacts (1 to 9) and the time period (8 to 128 seconds) within which the impacts must occur to generate an alarm signal, and the minimum time duration (1.0 to 5.0 seconds) a sustained attack must exceed to generate an alarm signal.

The detector shall, after initial setup, stabilize within five minutes after being powered on and after any alarm initiating stimulus is removed.

3.4.1 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature concerning performance data that would exceed the minimum requirements of this specification.

3.5 Enclosures. Where enclosures are applicable, they shall have a removable cover arranged to provide ready access for mounting, wiring, alignment, and for sensitivity and other internal adjustments. Mounting holes shall be provided in the back of the enclosure in positions that are readily accessible (sufficiently displaced from all internal parts and the enclosure side) when all internal parts and wiring are in place. No external hardware or fasteners except the cover fasteners shall be used.

3.6 Cover. The processor cover shall be designed such that any attempted access to the interior shall cause a tamper condition. The tamper circuits shall be in the secure state whenever the cover is in place and shall go to their alarm state when the cover is moved sufficiently to reveal the inside or to introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.7 Terminals. Screw terminals shall be provided for interconnecting wiring. Access to the terminals shall require removal of the enclosure cover.

3.8 Switch and/or relay assembly. Alarm and tamper switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the published current rating shall not cause damage to the switch or contacts.

3.9 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this

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damage from impacts which do not damage the enclosure. System

specification and shall be stated in the manufacturer's literature.

3.10 Maintainability. The flexible cable detector shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the device or replaceable sub-assembly and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the detector beyond mere removal of the cover.

3.11 Life cycle. The flexible cable detector alarm and tamper switches and/or relays shall be rated at not fewer than 1 000 000 operations under a load of 250 milliampere driven from a 48 volt DC source. Contact resistance shall not change more than +20% from the initial reading.

3.12 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Performance tests. Each of the following tests shall be conducted on wall surfaces constructed in a test facility for the purpose of evaluating flexible cable detectors. Three walls shall be constructed for these tests in the best commercial practice for wall construction. Each shall be at least 2400 mm high , at least 9100 mm long, and consist of:

- 12.7 mm sheetrock attached with screws to both sides of a 50 X 100 mm wood stud frame;
- 200 X 200 X 400 mm cinder blocks; and
- 600 X 2500 mm by a minimum of 0.038 cm. sheet metal attached with screws to both sides of a 50 X 100 mm steel stud frame.

Each wall shall have 9000 mm of 12.7 mm conduit attached to it at 910 mm intervals up 1220 mm from the floor.

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For each of the following tests, the processor's alarm contacts shall be connected to an electronic sounder, a light emitting diode, or an incandescent light bulb to indicate an alarm. The processor's sensitivity shall be adjusted to maximum.

4.1.1 Sheetrock surface penetration test.

4.1.1.1 Test with detector mounted directly to surface. In accordance with the manufacturer's recommendations, 9000 mm of the sensing cable is mounted directly to the surface of the sheetrock wall 910 mm above the floor.

(a) Low frequency band test. At the processor, the low frequency band is selected for monitoring; the number of impacts shall be set for 5; the impact time window shall be set for 38 seconds; and the sustained attack time window shall be set for 5 seconds.

On the side of the wall opposite the sensing cable and 610 mm above the floor, the wall shall be struck with a normal carpenter's claw hammer at a rate of one strike every three seconds. A breaking penetration of the wall shall be attempted. The amount of time and the number of strikes required to cause an alarm shall be recorded. An alarm shall be signaled within two seconds after the fifth strike is made. The detector shall be allowed to reset and the striking is resumed at a rate of one strike every 15 seconds for two minutes without an alarm being generated.

(b) High frequency band test. At the processor, the high frequency band shall be selected for monitoring. The other settings shall remain the same.

On the side of the wall opposite the sensing cable and 610 mm above the floor, the sheetrock is cut with a sheetrock or keyhole saw. The cutting shall be continued until an alarm is signaled. The amount of time required to generate an alarm shall be within $\pm$ 20% of the sustained attack time setting. The detector shall be allowed to reset and the cutting is resumed for 3 seconds without an alarm being generated.

4.1.1.2 Test with detector installed in conduit mounted to surface. The detector cable shall be installed to be tested in the conduit mounted on the sheetrock wall and two steps in 4.1.1.1 are repeated.

4.1.2 Cinder block surface penetration test.

4.1.2.1 Test with detector mounted directly to surface. In

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accordance with the manufacturer's recommendations, about 9000 mm of the sensing cable shall be mounted directly to the surface of the cinder block wall 910 mm above the floor.

(a) Low frequency band test. At the processor, the low frequency band shall be selected for monitoring; the number of impacts shall be set for 5; the impact time window shall be set for 38 seconds; and the sustained attack time window shall be set for 5 seconds.

On the side of the wall opposite the sensing cable and 610 mm above the floor, the wall shall be struck with a standard 5.45 kg long handled sledge hammer at a rate of one strike every three seconds. The amount of time and the number of strikes required to cause an alarm shall be recorded. An alarm shall be signaled within two seconds after the fifth strike. The detector shall be allowed to reset and the striking shall be resumed at a rate of one strike every 15 seconds for two minutes without an alarm being generated.

(b) High frequency band test. At the processor, the high frequency band shall be selected for monitoring. The other settings shall remain the same.

On the side of the wall opposite the sensing cable, the cinder block wall shall be drilled about 610 mm inches above the floor with a standard 12.7 mm cement bit and a normal 12.7 mm drill motor. Drilling shall be continued until an alarm is signaled. The amount of time required to generate an alarm shall be within ± 20% of the sustained attack time setting. The detector shall be allowed to reset and the drilling shall be resumed for 3 seconds, without an alarm being generated.

4.1.2.2. Test with detector installed in conduit mounted to surface. The detector cable to be tested shall be installed in the conduit mounted on the cinder block wall and two steps 4.1.2.1 are repeated.

4.1.3.1 Metal surface penetration test.

4.1.3.1 Test with detector mounted directly to surface. In accordance with the manufacturer's recommendations, about 9.1 m. of the sensing cable shall be mounted directly to the surface of the metal panel wall 91 cm. above the floor.

(a) Low frequency band test. At the processor, the low frequency band shall be selected for monitoring; the number of impacts shall be set for 5; the impact time window shall be set for 38 seconds; and the sustained attack time window shall be set for 5 seconds.

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On the side of the wall opposite the sensing cable and 610 mm above the floor, a hole shall be cut in the wall with a standard fire axe. The axe shall be struck at a rate of one strike every three seconds. The amount of time and the number of strikes required to cause an alarm shall be recorded. An alarm shall be signaled within two seconds after the fifth strike. The detector shall be allowed to reset and the striking shall be resumed at a rate of one strike every 15 seconds for two minutes without an alarm being generated.

(b) High frequency band test. At the processor, the high frequency band shall be selected for monitoring. The other settings shall remain the same.

On the side of the wall opposite the sensing cable and 610 mm above the floor, the metal panel shall be cut with a reciprocating saw with a metal cutting blade and a 50 mm steel hole saw driven by a 9.1 mm drill motor. Cutting shall be continued until an alarm is signaled. The amount of time required to generate an alarm shall be within ± 20% of the sustained attack time setting. The detector shall be allowed to reset and the cutting shall be resumed for 3 seconds without an alarm being generated.

4.1.3.2 Test with detector installed in conduit mounted to surface. The detector cable to be tested shall be installed in the conduit mounted on the metal panel wall and the two steps in 4.1.3.1 are repeated.

4.2 Stability test. The equipment submitted for qualification testing shall perform its intended function in a stable environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

4.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.4 Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0

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degrees C. The unit shall then be tested as specified in 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. A detector with 3000 mm of cable shall be tested for compliance with 3.12 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber. The detector and cable shall be mounted so as to simulate mounting on an interior wall.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.

4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil agency marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

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6. NOTES

6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration
Federal Supply Service, Furniture Commodity Center
Washington, DC. 20406

Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

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6.4.1 Alarms.

6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.1 Invalid alarm. Any other alarm.

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